

Experiential Learning: Statistics and Probability

1. Introduction to Experiential Learning

Experiential Learning (EL) integrates mathematical concepts with real-life engineering, technological, and societal problems. These instructions guide students to explore, apply, analyze, and reflect on Statistics and Probability concepts through active learning.

2. Activity Objectives

The Objective of this activity is to

- Apply Statistics and Probability concepts to real-world engineering or societal problems.
- Develop problem-solving, analytical, and decision-making skills.
- Understand the interdisciplinary role of Statistics in engineering.
- Enhance communication skills through structured documentation and presentation.

3. Step-by-Step Instructions for Students

1. Problem Identification: Select a real-world problem from engineering, environment, economics, data science, or any societal domain, which is relevant to the topic studies in the present semester

2. Context Analysis: Describe why this problem is important. Identify constraints, variables, and assumptions.

3. Data Collection or Simulation: Collect real data (measurements, datasets, surveys) or assume parameters with justification.

4. Mathematical Framing:

- For Statistics: Define data, processing of data, trends and patterns identification, testing the hypothesis and give inference.
- For Probability: Formulate Probabilistic models for the data, analyze the model, estimate the parameters and give inference.

5. Model Development: Convert the data collected for the societal problem into a statistical model using Statistics and Probability methods.

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6. **Computations and Analysis:** Show details of collecting the data, pre-processing the data, constructing a model, creating a hypothesis and testing the hypothesis.
7. **Visualization:** Present graphs, tables, data structures, and plots to illustrate findings.
8. **Interpretation:** Explain the Statistical results mean in the context of the problem. Relate results to the real-world problem.
9. **Reflection:** Discuss challenges faced, insights gained, and how Statistics and Probability enhanced understanding.
10. **Conclusion:** Summarize outcomes and possible extensions of the work.

4. Tools and Resources

- Choose appropriate tool such as MATLAB or Python practiced in the previous and present semester to solve the modeled problem and visualize.
- Other software tools such as Octave, Microsoft Excel and so on can also be used if required and learnt.
- Graphing tools (Desmos, GeoGebra, Matplotlib).
- Engineering case studies and open datasets.

5. Submission Guidelines

- Submit a typed report in the given template, preferable spiral binding.
- Include graphs, tables, and images clearly labeled.
- Attach raw data and calculations in an appendix.
- Maintain academic integrity and cite all sources.
- Limit report length to 5–12 pages.
- Last date for submission of the report and presentation: on or before 25-04-2026

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6. Evaluation Criteria and Rubric

Criteria	Excellent (20-25)	Good (15-19)	Satisfactory (10-14)	Needs Improvement (1-9)
1. Problem Identification & Context	Clearly identifies a real-world problem; strong justification and relevance	Problem is identified with some justification	Problem identified but justification weak	Problem unclear; little or no context
2. Data Collection and Analysis	Accurate and appropriate collection of data, analysis of data and graphical representation	Mostly correct techniques with minor errors	Basic techniques used; several errors	Incorrect or minimal use of data, analysis and representation
3. Statistical Modeling and Computations	Well-formulated model; detailed and accurate computations.	Good model; mostly correct calculations.	Basic model; several computational mistakes.	Model missing/incorrect; major errors.
4. Visual Representations and Interpretations (Graphs, Tables, explanations)	Clear, accurate, insightful visuals. Deep, meaningful interpretation linked to real context	Appropriate visuals with minor issues. Good interpretation; some linking	Basic visuals; limited clarity and interpretation	Missing or unclear visuals and very little interpretation
5. Reflection & Critical Thinking	Insightful reflection, highlights learning & challenges	Adequate reflection	Minimal reflection	No meaningful reflection
6. Presentation & Documentation Quality	Professional, well-structured, error-free	Good organization; few errors	Acceptable; some errors	Poorly organized; many errors
Total Score (out of 150)				